

BUNDABERG, Australia

WMO: 943870

Lat: 24.9069S

Lon: 152.3220E

Elev: 105

StdP: 14.64

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	45.7	48.2	27.4	21.4	62.4	31.3	25.7	61.7	22.4	67.0	20.2	65.8	5.4	230	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.6	89.0	74.4	87.4	73.9	86.1	73.2	78.3	84.1	77.3	83.0	76.3	82.1	13.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.8	140.0	81.3	75.7	134.9	80.7	74.6	129.9	79.9	41.9	84.1	40.9	83.0	39.9	82.1	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	20.3	18.4	40.7	94.6	1.6	2.2	39.5	96.2	38.6	97.5	37.7	98.7	36.5	100.3		
(5)			37.5	80.4	1.4	1.6	36.5	81.6	35.7	82.5	34.9	83.4	33.9	84.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.7	78.4	78.1	76.3	72.2	67.0	62.9	61.6	63.0	67.5	71.3	74.1	77.0
	DBStd	6.78	2.65	2.78	2.47	2.80	3.51	3.87	3.50	3.52	3.33	3.35	3.21	3.26
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	311	0	0	0	1	21	83	114	80	11	1	0	0
	CDD50	7569	881	786	816	667	528	386	359	403	525	659	722	837
	CDD65	2405	416	366	351	217	84	19	9	18	86	195	272	372
	CDH74	16469	3554	2963	2402	1073	281	38	29	87	411	983	1691	2957
	CDH80	3916	1054	879	576	134	9	0	0	2	32	115	311	804
(14) Wind	WSAvg	9.2	10.4	10.1	9.7	9.1	8.3	8.1	7.9	8.2	8.9	9.6	9.7	10.2
(15) Precipitation	PrecAvg	40.0	6.9	6.1	4.3	2.4	2.9	1.9	1.5	1.4	1.4	2.9	3.6	5.0
	PrecMax	74.5	27.3	28.7	17.0	10.9	15.0	14.1	15.0	5.3	5.7	9.6	8.9	18.9
	PrecMin	19.0	0.0	0.6	0.3	0.1	0.4	0.0	0.0	0.0	0.0	0.4	0.1	0.3
	PrecStd	12.9	5.5	5.4	3.8	2.2	2.8	2.6	2.3	1.3	1.4	2.2	2.2	4.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.6	91.3	89.2	85.7	81.1	77.3	76.9	79.0	83.7	85.5	88.2	91.5
		MCWB	76.7	75.9	74.2	71.4	66.5	65.5	62.4	63.3	65.0	67.2	69.0	71.2
	2%	DB	88.4	88.6	86.9	83.1	78.8	74.9	74.4	76.2	80.2	82.8	85.1	87.8
		MCWB	75.3	75.1	73.6	69.7	65.6	63.7	61.2	62.2	65.0	68.7	71.0	74.1
	5%	DB	86.9	86.9	85.1	81.5	77.1	73.2	72.5	74.4	78.2	80.9	83.4	85.9
		MCWB	74.3	74.3	72.6	68.7	64.9	62.8	60.5	61.0	64.4	68.2	70.2	72.9
	10%	DB	85.4	85.2	83.4	79.9	75.3	71.2	70.7	72.6	76.4	79.2	81.6	84.3
		MCWB	73.5	73.4	71.7	68.2	64.2	61.6	59.5	59.9	63.6	66.9	69.2	71.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.0	79.0	78.4	76.1	72.2	69.1	68.5	68.2	71.5	73.8	76.2	79.0
		MCDB	85.9	85.4	83.0	80.5	76.3	72.6	72.5	74.3	77.0	79.8	82.2	85.4
	2%	WB	78.4	77.9	76.7	73.4	70.4	67.1	66.0	66.3	69.4	72.5	74.4	77.3
		MCDB	84.2	83.9	81.9	78.3	74.3	70.9	70.0	71.8	75.6	78.2	80.7	83.2
	5%	WB	77.4	76.9	75.5	71.7	68.3	65.5	63.9	64.6	68.1	71.1	72.9	76.0
		MCDB	83.0	82.9	81.1	77.7	73.7	70.0	68.4	70.1	73.7	77.0	79.4	82.0
	10%	WB	76.2	75.8	74.3	70.3	66.6	63.9	62.0	63.0	66.9	69.8	71.7	74.7
		MCDB	82.0	81.8	80.0	76.7	72.5	69.2	67.6	68.9	72.5	75.7	78.2	80.8
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	13.6	14.3	15.7	17.3	17.1	19.3	20.2	18.7	16.1	14.9	14.1
		MCDBR	15.2	15.8	16.4	17.3	18.8	18.2	20.8	21.8	20.5	17.4	16.4	15.4
	5% WB	MCWBR	4.9	5.3	5.5	6.5	8.1	8.6	9.9	10.0	8.5	6.6	5.9	5.4
		MCWBR	12.7	13.1	13.1	14.3	15.0	14.7	15.8	17.4	15.9	14.0	13.9	12.7
(40) Clear-Sky Solar Irradiance	taub		0.391	0.384	0.371	0.343	0.322	0.315	0.306	0.318	0.352	0.376	0.378	0.392
		taud	2.513	2.543	2.567	2.635	2.639	2.653	2.656	2.609	2.506	2.470	2.499	2.491
	Ebn at Noon		302	300	295	290	284	279	286	294	296	298	304	302
		Edh at Noon	36	34	32	28	25	23	24	28	33	36	36	37
(44) All-Sky Solar Radiation	RadAvg		2105	1920	1695	1490	1251	1075	1220	1503	1826	1999	2152	2153
	RadStd		152	179	141	114	64	83	97	107	134	155	201	220

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212	+193
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page